

FIITJEE – Phase Test -4 (JEE-

PHYSICS, CHEMISTRY & MATHEMATICS

CPT1-2

Q.P.CODE:0000

PAPER - 2

13/08/2023

Time Allotted: 3 Hours

Maximum Marks: 183

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. All the section can be filled in **PART-A** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For Only One Part.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened. **Negative Marks: -1** In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-1 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-14)** – Contains seven (07) multiple choice questions which have ONLY ONE CORRECT answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iii) **Part-A (15-18)** - This section contains Two paragraphs. Based on each paragraph, there are Two multiple choice questions. Each question has only one correct answer and carries **+3 marks** for the correct answer. **There is no negative marking.**

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

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BATCHES – CTY 2224 X LOT (XII)

Useful Data Chemistry:

Gas Constant	R	=	8.314 J K ⁻¹ mol ⁻¹
		=	0.0821 Lit atm K ⁻¹ mol ⁻¹
		=	1.987 ≈ 2 Cal K ⁻¹ mol ⁻¹
Avogadro's Number	N _a	=	6.023 × 10 ²³
Planck's Constant	h	=	6.626 × 10 ⁻³⁴ Js
		=	6.25 × 10 ⁻²⁷ erg.s
1 Faraday		=	96500 Coulomb
1 calorie		=	4.2 Joule
1 amu		=	1.66 × 10 ⁻²⁷ kg
1 eV		=	1.6 × 10 ⁻¹⁹ J
Atomic No :	H=1, D=1, Li=3, Na=11, K=19, Rb=37, Cs=55, F=9, Ca=20, He=2, O=8, Au=79.		
Atomic Masses:	He=4, Mg=24, C=12, O=16, N=14, P=31, Br=80, Cu=63.5, Fe=56, Mn=55, Si = 28 Pb=207, u=197, Ag=108, F=19, H=2, Cl=35.5		

Useful Data Physics:

Acceleration due to gravity $g = 10 \text{ m/s}^2$

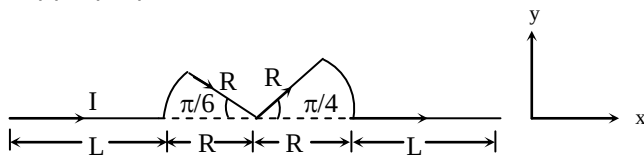


Section – I (Physics)

PART – A

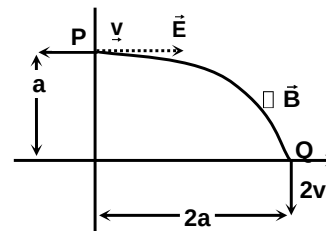
One Or More Than One Correct Type

1. A conductor (shown in the figure) carrying constant current I is kept in the x - y plane in a uniform magnetic field $\vec{B}$. If F is the magnitude of the total magnetic force acting on the conductor, then the correct statement(s) is(are)

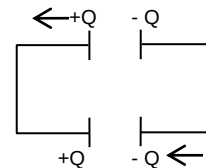


- (A) If $\vec{B}$ is along $\hat{z}$, $F \propto (L + R)$ (B) If $\vec{B}$ is along $\hat{x}$, $F = 0$
 (C) If $\vec{B}$ is along $\hat{y}$, $F \propto (L + R)$ (D) If $\vec{B}$ is along $\hat{z}$, $F = 0$
2. When a current-carrying coil is placed in a uniform magnetic field with its magnetic moment anti-parallel to the field.
 (A) Torque on it is maximum (B) Torque on it is zero
 (C) Potential energy is maximum (D) Dipole is in unstable equilibrium.
3. Two identical charged particles enter a uniform magnetic field with same speed but at angles 30° and 60° with the field. Let a , b and c be the ratios of their time periods, radii and pitches of the helical paths. Then
 (A) $abc = 1$ (B) $abc > 1$ (C) $abc < 1$ (D) $a = bc$

4. A particle of charge $+q$ and mass m moving under the influence of a uniform electric field $E\hat{i}$ and uniform magnetic field $B\hat{k}$ follows a trajectory from P to Q as shown in figure. The velocities at P and Q are $v\hat{i}$ and $-2v\hat{j}$ which of the following statement (s) is/are correct?

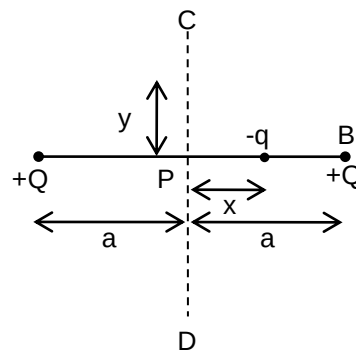


- (A) $E = \frac{3}{4} \left[\frac{mv^2}{qa} \right]$
 (B) Rate of work done by the electric field at P is $\frac{3}{4} \left[\frac{mv^3}{a} \right]$
 (C) Rate of work done by the electric field at P is zero
 (D) Rate of work done both the fields at Q is zero
5. A charge Q is imparted to two each identical capacitors in parallel. Separation of the plates in each capacitor is d_0 . Suddenly, the first plate of the first capacitor and the second plate of the second capacitor start moving to the left with speed v , then



- (A) charges on the two capacitors as a function of time are $\frac{Q(d_0 - vt)}{d_0}$, $\frac{Q(d_0 + vt)}{d_0}$
 (B) charges on the two capacitors as a function of time are $\frac{Qd_0}{(d_0 - vt)}$, $\frac{Qd_0}{(d_0 + vt)}$
 (C) current in the circuit will increase as time passes on
 (D) current in the circuit will be constant

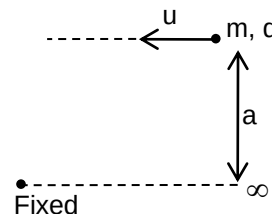
6. For the arrangement shown in Fig. the two positive charges, $+Q$ each, are fixed. Mark out the correct statement(s) regarding a third charged particle $-q$ placed at mid-point P that can be displaced along or perpendicular to the line connecting the charges.
- (A) The particle will perform S.H.M. for $x \ll a$
 (B) The particle will oscillate about P but not harmonically for any x .
 (C) The particle will perform S.H.M. for $y \ll a$
 (D) The particle will oscillate about P but not harmonically for y comparable to a .



7. A charge particle q is projected in an electric field produced by a fixed point charge Q as shown in Fig. Mark out the correct statements.
- (A) The path taken by q is a straight line
 (B) The path taken by q is not a straight line.
 (C) The minimum distance between the two particles is

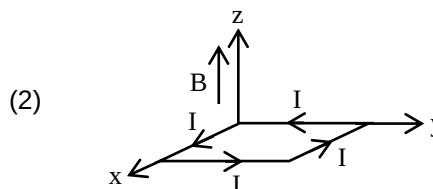
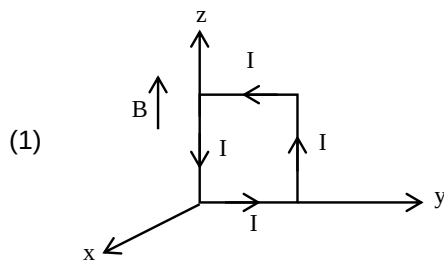
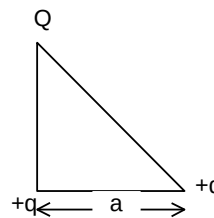
$$\frac{\frac{qQ}{2\pi\epsilon_0} + \sqrt{\left(\frac{qQ}{2\pi\epsilon_0}\right)^2 + 4m^2u^2a^2}}{2mu^2}$$

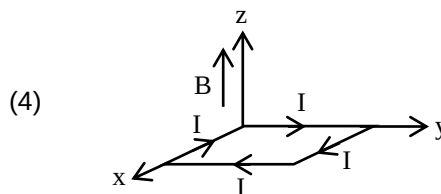
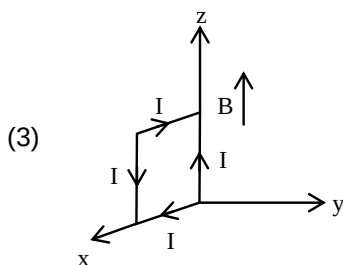
- (D) Velocity of the particle q is changing in both magnitude and direction.



Single Correct Choice Type

8. In an experiment, electrons are accelerated, from rest, by applying, a voltage of 500 V. Calculate the radius of the path if a magnetic field 100 mT is then applied. [Charge of the electron = 1.6×10^{-19} C
Mass of the electron = 9×10^{-31} kg]
 (A) 7.5×10^{-3} m (B) 7.5×10^{-2} m (C) 7.5 m (D) 7.5×10^{-4} m
9. Three charges Q , $+q$ and $+q$ are placed at the vertices of a right-angled isosceles triangle as shown. The net electrostatic energy of the configuration is zero if Q is equal to
 (A) $\frac{-q}{1+\sqrt{2}}$ (B) $\frac{-2q}{2+\sqrt{2}}$
 (C) $-2q$ (D) $+q$
10. A rectangular loop of sides 10 cm and 5 cm carrying a current I of 12 A is placed in different orientations as shown in the figures below:



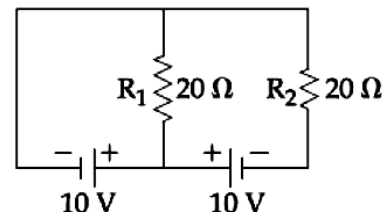


If there is a uniform magnetic field of 0.3 T in the positive z direction, in which orientations the loop would be in (i) stable equilibrium and (ii) unstable equilibrium?

- (A) (1) and (3), respectively
(B) (2) and (4), respectively
(C) (2) and (3), respectively
(D) (1) and (2), respectively

11. In the given circuit the cells have zero internal resistance. The currents (in Amperes) passing through resistance R_1 and R_2 respectively, are:

- (A) 1, 2
(B) 2, 2
(C) 0.5, 0
(D) 0, 1

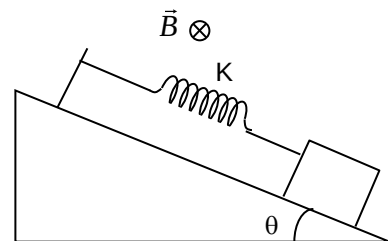


12. Two identical thin rings, each of radius R metres, are coaxially placed a distance r metres apart. If Q_1 coulomb and Q_2 coulomb are, respectively, the charges uniformly spread on the two rings, the work done in moving a charge q from the centre of one ring to that of the other is

- (A) zero
(B) $\frac{q(Q_1 - Q_2)(\sqrt{2} - 1)}{(4\sqrt{2}\pi\epsilon_0 R)}$
(C) $\frac{q\sqrt{2}(Q_1 + Q_2)}{(4\pi\epsilon_0 R)}$
(D) $\frac{q(Q_1 + Q_2)(\sqrt{2} + 1)}{(4\sqrt{2}\pi\epsilon_0 R)}$

13. A small block of mass m , having charge q , is placed on a frictionless inclined plane making an angle θ with the horizontal. There exists a uniform magnetic field B parallel to the inclined plane but perpendicular to the length of spring. If m is slightly pulled on the incline in downward direction, the time period of oscillation will be (assume that the block does not leave contact with the plane)

- (A) $2\pi\sqrt{\frac{m}{K}}$
(B) $2\pi\sqrt{\frac{2m}{K}}$
(C) $2\pi\sqrt{\frac{qB}{K}}$
(D) $2\pi\sqrt{\frac{qB}{2K}}$

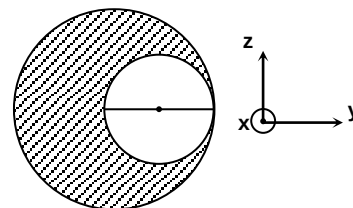


14. A parallel plate capacitor is made by stacking n equally spaced plates connected alternatively. If the capacitance between any two adjacent plates is C , then the resultant capacitance is

- (A) nC
(B) C
(C) $(n+1)C$
(D) $(n-1)C$

Comprehension Type
Paragraph for question Nos. 15 to 16

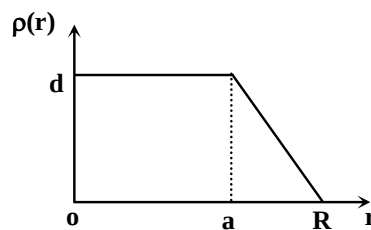
A long cylindrical conductor of radius R having a cylindrical cavity of radius $R/2$ through out its length is lying with its axis parallel to x -axis. The distance between the axes of the cylinder and the cavity is $R/2$. A constant current I is passed through the cylinder along its length in the positive x direction. The cross-sectional view of the cylinder is shown in the figure.



15. The magnetic induction at any point inside the cavity is
 (A) $\frac{\mu_0 I}{3\pi R}(-\hat{k})$ (B) $\frac{\mu_0 I}{3\pi R}(\hat{k})$ (C) $\frac{\mu_0 I}{6\pi R}(\hat{j})$ (D) $\frac{\mu_0 I}{6\pi R}(-\hat{j})$
16. A charged particle having specific charge (charge to mass ratio) is $\frac{18\pi}{\mu_0 I} \times 10^6$ SI unit is projected from a point on the axis of the cavity along x -axis with the speed of 10^6 m/s. Radius of curvature of the path of the particle is equal to
 (A) $R/18$ (B) $R/9$ (C) $R/6$ (D) $R/3$

Paragraph for Question Nos. 17 to 18

The nuclear charge (Ze) is non-uniformly distributed within a nucleus of radius R . The charge density $\rho(r)$ [charge per unit volume] is dependent only on the radial distance r from the centre of the nucleus as shown in figure. The electric field is only along the radial direction.



17. For $a = 0$, the value of d (maximum value of ρ as shown in the figure) is
 (A) $\frac{3Ze}{4\pi R^3}$ (B) $\frac{3Ze}{\pi R^3}$ (C) $\frac{4Ze}{3\pi R^3}$ (D) $\frac{Ze}{3\pi R^3}$
18. The electric field within the nucleus is generally observed to be linearly dependent on r . This implies.
 (A) $a = 0$ (B) $a = \frac{R}{2}$ (C) $a = R$ (D) $a = \frac{2R}{3}$

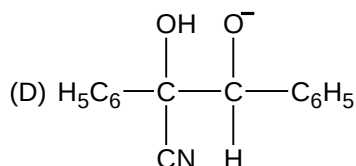
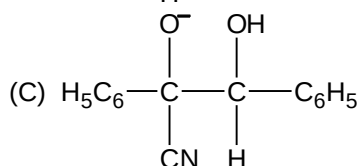
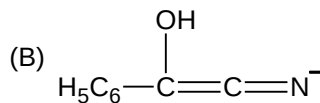
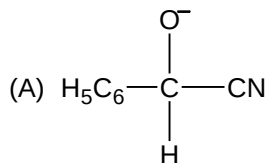
space for rough work

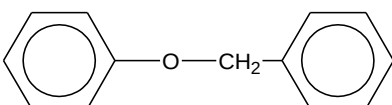
Section – II (Chemistry)

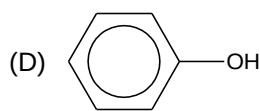
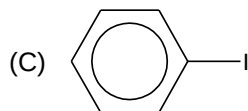
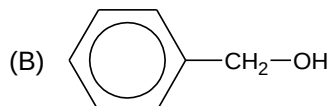
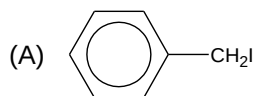
PART – A

One or More Than One Correct Type

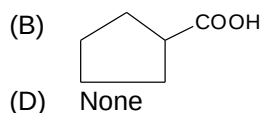
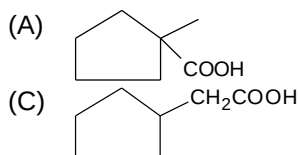
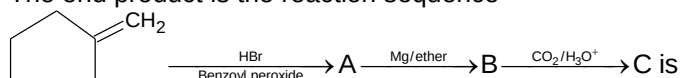
- Mixture of $\text{C}_6\text{H}_5\text{CHO}$ and HCHO is treated with NaOH then Cannizzaro reaction involves
 (A) oxidation of HCHO (B) reduction of HCHO
 (C) oxidation of $\text{C}_6\text{H}_5\text{CHO}$ (D) reduction of $\text{C}_6\text{H}_5\text{CHO}$
- In benzoin condensation which of the following are formed as intermediate compounds?



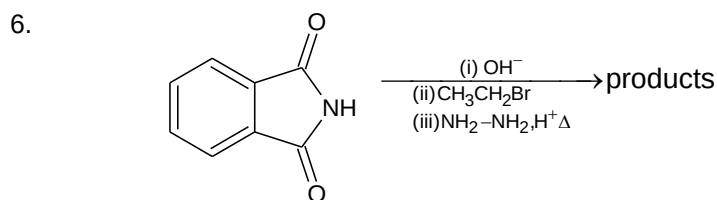
- The ether  when treated with HI produces,



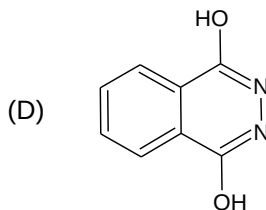
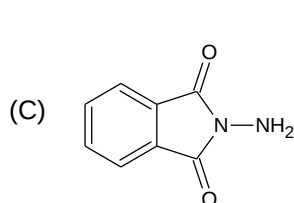
- The end product is the reaction sequence



- Primary amine can be formed by the following process?
 (A) Schmidt reaction (B) Gabriel phthalimide synthesis
 (C) Hoffmann bromamide degradation (D) Carbyl amine reaction



- Product are
 (A) $\text{CH}_3\text{CH}_2\text{NH}-\text{NH}_2$ (B) $\text{CH}_3\text{CH}_2\text{NH}_2$

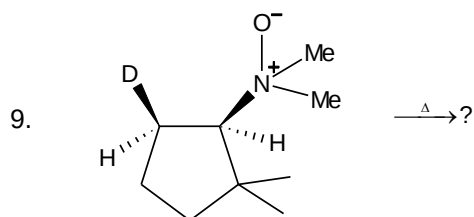
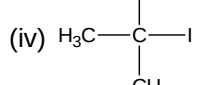
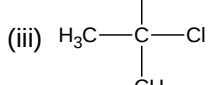
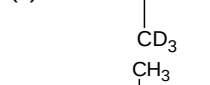
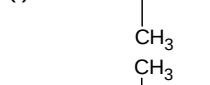
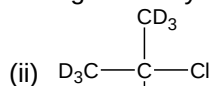
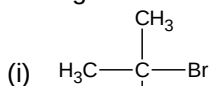


7. Benzyl chloride (PhCH_2Cl) can be prepared from Toluene by
 (A) SO_2Cl_2 (B) SOCl_2 (C) $\text{Cl}_2/h\nu$ (D) NaOCl

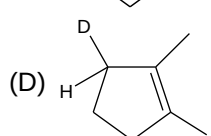
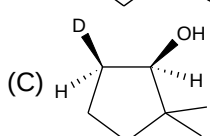
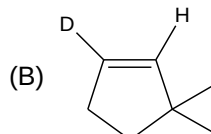
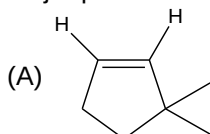
PART – A: (Single Correct Answer Type)

This section contains **07 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

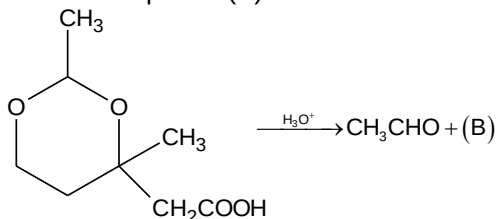
8. Arrange the following compound in order of decreasing reactivity in the E2 reaction:



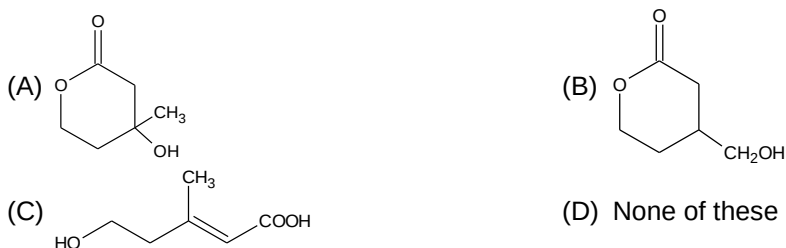
Major product of the above reaction is



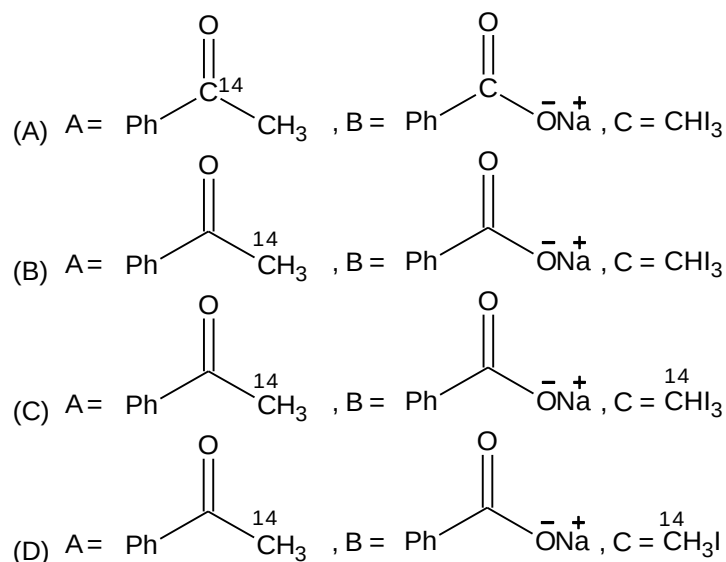
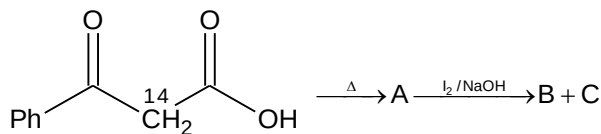
10. On standing in dilute aqueous acid, the compound (A) is smoothly converted into CH_3CHO and a neutral compound (B).



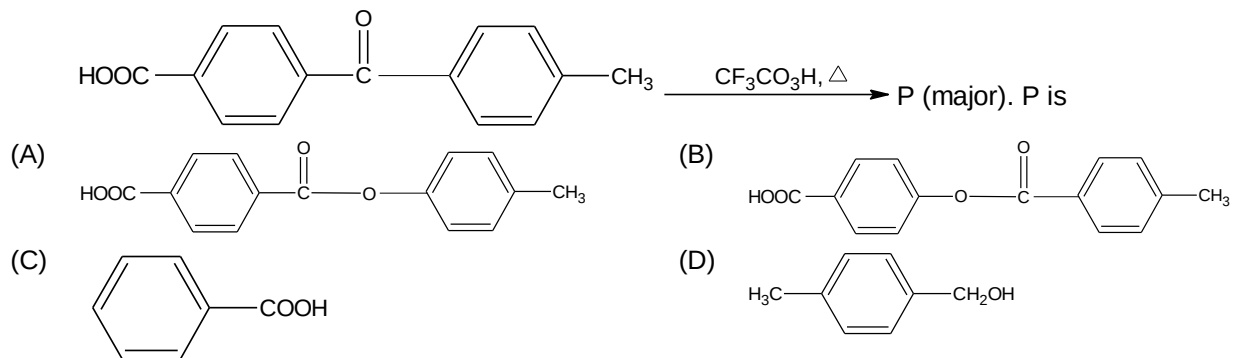
The compound (B) is



11.



12.

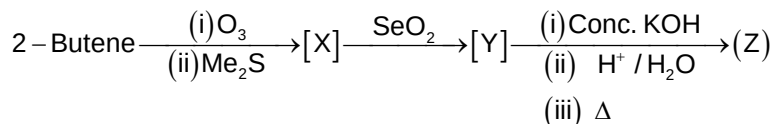


13.

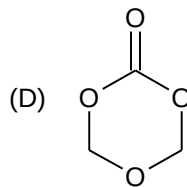
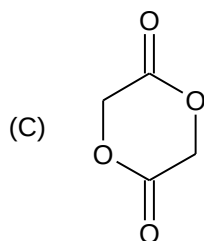
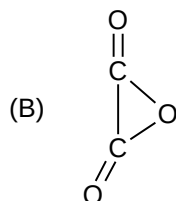
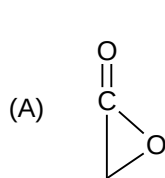
Decolourization of Br_2 would occur if which of the following compounds is treated with $\text{Br}_2 / \text{H}_2\text{O}$?

- (A) D-Glucose (B) D-Fructose
(C) both of these (D) none of these

14.



The end product [Z] is

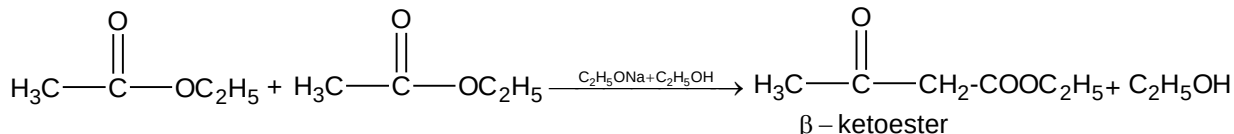


PART – A: (Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

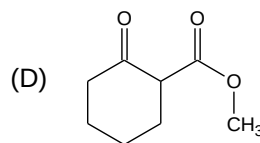
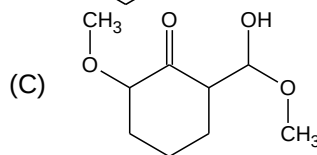
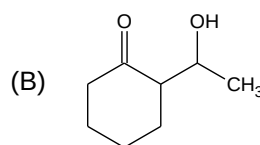
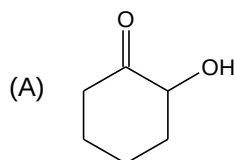
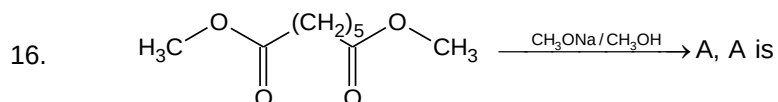
Paragraph for Question Nos. 15 to 16

The reaction takes place between two molecules of esters (same or different). One ester should have at least one α -hydrogen atom which behaves as a reagent. The reaction is catalysed by strong base like $\text{C}_2\text{H}_5\text{O}^-\text{Na}^+$ in presence of $\text{C}_2\text{H}_5\text{OH}$. The product of this reaction is β -keto ester.



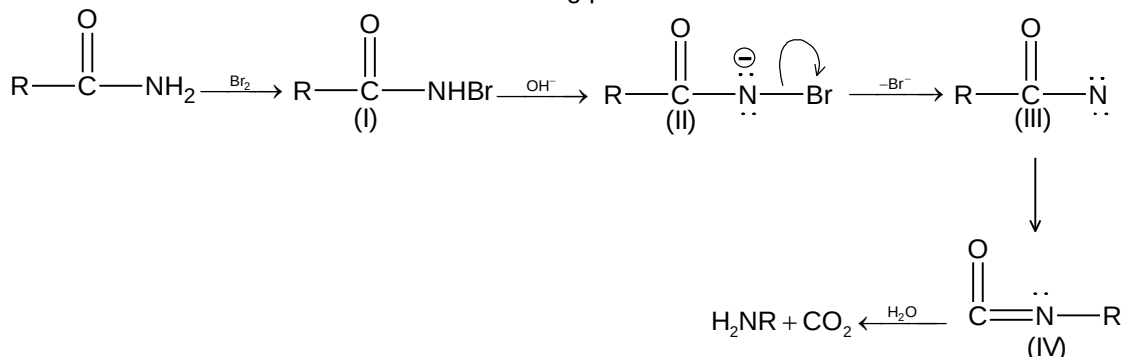
15. Which of the following esters will undergo Claisen condensation to form acetoacetic ester?

- (A) $\text{CH}_3\text{COOC}_2\text{H}_5$ (B) $\text{C}_6\text{H}_5\text{COOC}_2\text{H}_5$
(C) HCOOC_2H_5 (D) None of these



Paragraph for Question Nos. 17 to 18

Hofmann reaction is found to follow the following path:



17. Hofmann reaction, when carried out in presence of methanol containing sodium methoxide, instead of NaOH, then $\text{C}_6\text{H}_5\text{CONH}_2 \xrightarrow[\text{CH}_3\text{OH}]{\text{Br}_2, \text{CH}_3\text{ONa}}$ Product is

(A) CH_3NHCH_3 (B) $\text{C}_6\text{H}_5\text{NHCOOCH}_3$ (C) $\text{C}_6\text{H}_5\text{NHCH}_3$ (D) $\text{C}_6\text{H}_5\text{NHBr}$

18. $\text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{ND}_2 \xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) Br}_2, \text{NaOH}}$ Amine, Z is

(A) CH_3-ND_2 (B) CH_3NH_2
 (C) mixture of (A) and (B) (D) Reaction not possible

space for rough work

Section – III (Mathematics)

PART – A

One or More Than One Correct Type

1. $f : R \rightarrow R$ Such that for all $x, y \in R$, $|f(x) - f(y)| \leq |x - y|^3$ Then
 (A) f is differentiable for all $x \in R$ (B) f is continuous for all $x \in R$
 (C) $f(x) = c$ (constant for all $x \in R$) (D) Range of ' f ' is a singleton
2. A continuous and differentiable function ' f ' satisfies the condition ,
 $\int_0^x f(t)dt = f^2(x) - 1$ for all real ' x '. Then :
 (A) $f(x)$ is monotonic increasing $\forall x \in R$
 (B) $f(x)$ is monotonic decreasing $\forall x \in R$
 (C) $f(x)$ has no global extremum
 (D) the graph of $y = f(x)$ is a straight line.
3. Let $f(x) = \int_0^x \frac{\sin t}{t} dt (x > 0)$ then $f(x)$ has :
 (A) Local maxima if $x = n\pi$ where $n = 1, 3, 5, \dots$
 (B) Local minima if $x = n\pi$ where $n = 2, 4, 6, \dots$
 (C) Local maxima if $x = n\pi$ where $n = 2, 4, 6, \dots$
 (D) The function is monotonic
4. The value of $\int_0^1 \frac{2x^2 + 3x + 3}{(x+1)(x^2 + 2x + 2)} dx$ is :
 (A) $\frac{\pi}{4} + 2\ln 2 - \tan^{-1} 2$ (B) $\frac{\pi}{4} + 2\ln 2 - \tan^{-1} \frac{1}{3}$
 (C) $2\ln 2 - \cot^{-1} 3$ (D) $-\frac{\pi}{4} + \ln 4 + \cot^{-1} 2$
5. The value of $\int_0^\infty \frac{dx}{1+x^4}$ is
 (A) same as that of $\int_0^\infty \frac{x^2 + 1 dx}{1+x^4}$ (B) $\frac{\pi}{2\sqrt{2}}$
 (C) same as that of $\int_0^\infty \frac{x^2 dx}{1+x^4}$ (D) $\frac{\pi}{\sqrt{2}}$
6. The extremum values of the function $f(x) = \frac{1}{\sin x + 4} - \frac{1}{\cos x - 4}$, where $x \in R$ is :
 (A) $\frac{4}{8-\sqrt{2}}$ (B) $\frac{2\sqrt{2}}{8-\sqrt{2}}$ (C) $\frac{2\sqrt{2}}{4\sqrt{2}+1}$ (D) $\frac{4\sqrt{2}}{8+\sqrt{2}}$
7. The function $f(x) = x^{\frac{1}{\ln x}}$
 (A) is a constant function (B) has a domain $(0,1) \cup (1,\infty)$
 (C) is such that $\lim_{x \rightarrow 1} f(x)$ exist (D) is a periodic function

Single Correct Choice Type

8. $\int_0^1 \frac{\tan^{-1} x}{x} dx$ is equal to
 (A) $\int_0^{\pi/4} \frac{\sin x}{x} dx$ (B) $\int_0^{\pi/2} \frac{x}{\sin x} dx$ (C) $\frac{1}{2} \int_0^{\pi/2} \frac{x}{\sin x} dx$ (D) $\frac{1}{2} \int_0^{\pi/4} \frac{x}{\sin x} dx$
9. The value of the function $f(x) = 1 + x + \int_1^x (\ln^2 t + 2 \ln t) dt$ where $f'(x)$ vanishes is :
 (A) e^{-1} (B) 0 (C) $2e^{-1}$ (D) $1 + 2e^{-1}$
10. If the function $f(x) = \frac{t + 3x - x^2}{x - 4}$, where 't' is a parameter has a local minimum and a local maximum in its domain then the range of values of 't' is
 (A) (0, 4) (B) (0, ∞) (C) ($-\infty$, 4) (D) (4, ∞)
11. Let $f(x) = e^{\{e^{|x|} \operatorname{sgn} x\}}$ and $g(x) = e^{[e^{|x|} \operatorname{sgn} x]}$, $x \in R$ where $\{x\}$ and $[]$ denotes the fractional part and integral part functions respectively. Also $h(x) = \ln(f(x)) + \ln(g(x))$ then for all real x, h(x) is
 (A) an odd function (B) an even function
 (C) neither an odd nor an even function (D) both odd as well as even function
12. $\int e^x \left(\tan^{-1} x + \frac{2x}{(1+x^2)^2} \right) dx$ is equal to
 (A) $e^x \left(\tan^{-1} x - \frac{1}{1+x^2} \right) + C$ (B) $e^x \left(\tan^{-1} x + \frac{1}{1+x^2} \right) + C$
 (C) $e^x \left(\cot^{-1} x - \frac{1}{1+x^2} \right) + C$ (D) $e^x \left(\tan^{-1} x + \frac{2}{1+x^2} \right) + C$
13. Let $f(x) = \begin{cases} \frac{x^2 - 4}{x - \sqrt{3x - 2}} & x < 2 \\ \frac{2^x + 2^{3-x} - 6}{\sqrt{2^{-x}} - 2^{1-x}} & x > 2 \end{cases}$, then
 (A) $f(2) = 8 \Rightarrow f$ is continuous at $x = 2$ (B) $f(2) = 16 \Rightarrow f$ is continuous at $x = 2$
 (C) $f(2^-) \neq f(2^+) \Rightarrow f$ is discontinuous (D) f has a removable discontinuity at $x = 2$
14. If y is a function of x then $\frac{d^2 y}{dx^2} + y \frac{dy}{dx} = 0$. If x is a function of y then the equation becomes
 (A) $\frac{d^2 x}{dy^2} + x \frac{dx}{dy} = 0$ (B) $\frac{d^2 x}{dy^2} + y \left(\frac{dx}{dy} \right)^3 = 0$
 (C) $\frac{d^2 x}{dy^2} - y \left(\frac{dx}{dy} \right)^2 = 0$ (D) $\frac{d^2 x}{dy^2} - x \left(\frac{dx}{dy} \right)^2 = 0$

Comprehension Type
Paragraph for question Nos. 15 to 16

Let us consider the integrals of the following forms $f\left(x, \sqrt{mx^2 + nx + p}\right)^{1/2}$

Case 1 : If $m > 0$, then put $\sqrt{mx^2 + nx + p} = u \pm x\sqrt{m}$

Case 2 : If $p > 0$, then put $\sqrt{mx^2 + nx + p} = ux \pm \sqrt{p}$

Case 3 : If quadratic equation $mx^2 + nx + p = 0$ has real roots α and β there put

$$\sqrt{mx^2 + nx + p} = (x - \alpha)u \text{ or } (x - \beta)u$$

15. If $I = \int \frac{dx}{x - \sqrt{9x^2 + 4x + 6}}$ to evaluate I, one of the most proper substitution could be

(A) $\sqrt{9x^2 + 4x + 6} = u \pm 3x$

(B) $\sqrt{9x^2 + 4x + 6} = 3u \pm x$

(C) $x = \frac{1}{t}$

(D) $9x^2 + 4x + 6 = \frac{1}{t}$

16. $\int \frac{(x + \sqrt{1+x^2})^{15}}{\sqrt{1+x^2}} dx$ is equal to

(A) $\frac{(x + \sqrt{1+x^2})^{16}}{10} + c$ (B) $\frac{1}{15(\sqrt{1+x^2} + x)} + c$ (C) $\frac{15}{(\sqrt{1+x^2} - x)} + c$ (D) $\frac{(x + \sqrt{1+x^2})^{15}}{15} + c$

Paragraph for question Nos. 17 to 18

Consider two functions $f(x) = \lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} \right)^n$ and $g(x) = -x^{4b}$,

where $b = \lim_{x \rightarrow \infty} (\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1})$. Then

17. $f(x)$ is

(A) e^{-x^2}

(B) $e^{-x^2/2}$

(C) e^{x^2}

(D) $e^{x^2/2}$

18. $f(g(x))$ is

(A) $e^{-x^2/2}$

(B) $e^{\frac{x^4}{2}}$

(C) $e^{x^2/2}$

(D) $e^{\frac{x^4}{2}}$

space for rough work

FIITJEE COMMON TEST

BATCHES: CTY 2224 X LOT
CLASS- XII
PHASE TEST – 1 (13-08-2023)
QP. Code: 00000
JEE ADVANCED PAPER-2
ANSWER KEY

SECTION - I: PHYSICS PART – A

- | | | | |
|------------|------------|------------|------------|
| 1. A, B, C | 2. B, C, D | 3. A, D | 4. A, B, D |
| 5. A, D | 6. C, D | 7. B, C, D | 8. D |
| 9. B | 10. B | 11. C | 12. B |
| 13. A | 14. D | 15. B | 16. C |
| 17. B | 18. C | | |

SECTION - II: CHEMISTRY PART – A

- | | | | |
|------------|---------------|---------|-------|
| 1. A, D | 2. A, B, C, D | 3. A, D | 4. C |
| 5. A, B, C | 6. B, D | 7. A, C | 8. B |
| 9. A | 10. A | 11. C | 12. A |
| 13. A | 14. C | 15. A | 16. D |
| 17. B | 18. B | | |

SECTION - III: MATHEMATICS PART – A

- | | | | |
|---------------|------------|------------|------------|
| 1. A, B, C, D | 2. A, C, D | 3. A, B | 4. A, C, D |
| 5. B, C | 6. A, C | 7. A, B, C | 8. C |
| 9. D | 10. C | 11. A | 12. A |
| 13. C | 14. C | 15. A | 16. D |
| 17. B | 18. D | | |

HINTS & SOLUTIONS

SECTION - I: PHYSICS

PART – A

1. A, B, C
2. B, C, D
3. A, D
4. A, B, D

Sol. In moving from point P to Q change in kinetic energy is given by $= \frac{1}{2}m(2v)^2 - \frac{1}{2}mv^2 = \frac{3}{2}mv^2$

The total work done in the process of going from P to Q by the electric field is $= \vec{F} \cdot \vec{d}$ where $\vec{d}$ is displacement traveled. So work done $= qE\hat{i} \cdot 2a\hat{i} = 2aqE$. Now we know that change in K.E = W.D $\Rightarrow$

$$\frac{3}{2}mv^2 = 2aqE$$

$$E = \frac{3}{4} \left[\frac{mv^2}{qa} \right]$$

Thus (A) is correct.

Now rate of work done by electric field at P is equal to $\vec{F} \cdot \vec{v} = qE \cdot v = \frac{3qmv^2v}{4qa} = \frac{3}{4} \left[\frac{mv^3}{a} \right]$

Hence, (B) is the correct choice.

Now rate of work done by magnetic field at Q = 0.

Rate of work done by electric field at Q = $qE \cdot (2v) \cdot \cos\left(\frac{\pi}{2}\right) = 0$

Thus (D) is correct

5. A, D
6. C, D
7. B, C, D
8. D

Sol. $k_e = \frac{p^2}{2M_e} = 500 e \quad \dots(i)$

$$\& R = \frac{p}{eB} = \frac{1000 m_e e}{eB} = \frac{1010 \times 9 \times 10^{-31}}{1.6 \times 10^{-19}} = 100 \times 7.541 \times 10^{-6}$$

9. B

Sol. $Qq/4\pi\epsilon_0 a + Qq/4\pi\epsilon_0 \sqrt{2}a + q^2/4\pi\epsilon_0 a = 0$
 $\Rightarrow Q = -2q/(2 + \sqrt{2})$.

10. B

Sol. Since $\vec{B}$ is uniform, only torque acts on a current carrying loop. $\vec{\tau} = (\vec{IA}) \times \vec{B}$

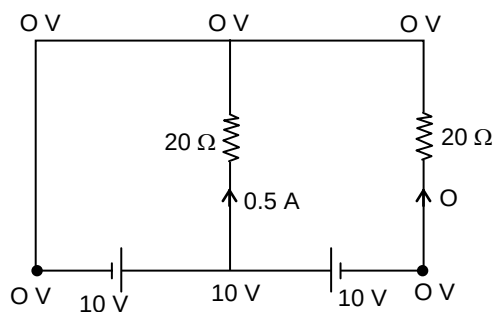
$\vec{A} = A\hat{k}$ for (2) and $\vec{A} = -A\hat{k}$ for (4).

$\therefore \vec{\tau} = 0$ for both these cases.

The energy of the loop in the $\vec{B}$ field is : $U = -\vec{IA} \cdot \vec{B}$, which is minimum for (2).

11. C

Sol.



12.

B

13.

A

14.

D

15.

B

16.

C

17.

B

Sol.

$$q = \int_0^R \frac{d}{R} (R-x) 4\pi x^2 dx = Ze$$

$$d = \frac{3Ze}{\pi R^3}$$

18.

C

Sol.

If within a sphere ρ is constant $E \propto r$

SECTION - II: CHEMISTRY

PART - A

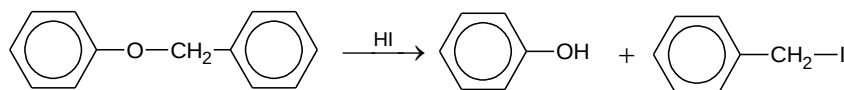
1.

Formaldehyde oxidises to formic acid and benzaldehyde oxidises to 'benzylic alcohol.'

2.

Check text for Benzoin condensation.

3.



6.

Reaction goes via Gabriel phthalimide synthesis.

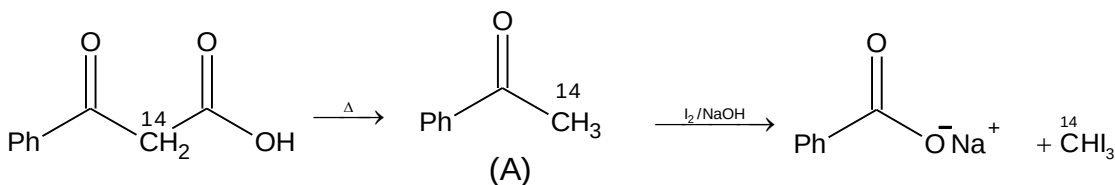
8.

 E_2 reaction mechanism depends on kinetic isotopic effect, good leaving group.

10.

Acetal group formation is reversible.

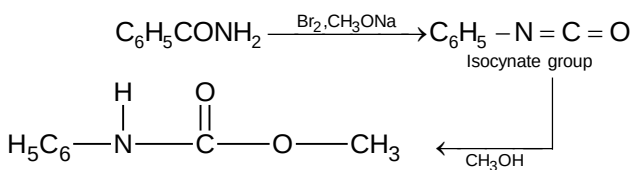
11.



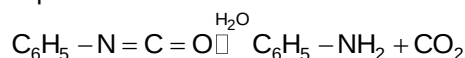
12.

Check Baeyer-Villiger oxidation.

17.



In presence of



SECTION- III: MATHEMATICS

PART – A

1. A, B, C, D

Sol.
$$0 \leq \lim_{x \rightarrow y} \left| \frac{f(x) - f(y)}{x - y} \right| \leq \lim_{x \rightarrow 0} |x - y|^2 = 0$$

$$\Rightarrow f'(x) = 0 \forall x \in R$$

$\therefore f$ is differentiable for all $x \in R$

$\Rightarrow f$ is continuous for all $x \in R$

$\Rightarrow f(x)$ is a constant

$\Rightarrow$ Range 'f' contains one element

2. A, C, D

Sol. Differentiating $f(x) = 2f(x) \cdot f'(x) \Rightarrow f'(x) = \frac{1}{2}; f(x) \neq 0$

Hence $f(x) = \frac{x}{2} + c$. Put $x=0$; $f(0)=c$; but $f^2(0)=1$

$$\Rightarrow f(0) = \pm 1$$

Hence $f(x) = \frac{x}{2} \pm 1$

3. A, B

Sol. $f'(x) = \frac{\sin x}{x}; f''(x) = \frac{x \cos x - \sin x}{x^2}$

Now $f''(2m\pi)$ we have $f''(x) > 0$ & $f''((2m-1)\pi)$ we have $f''(x) < 0$

4. A, C, D

Sol. Numerator $= 2(x^2 + 2x + 2) - (x + 1)$

5. B, C

Sol.

$$I = \int_0^{\infty} \frac{x^2}{1+x^4} dx + \int_0^{\infty} \frac{1-x^2}{1+x^4} dx = I_1 + I_2$$

$$I_2 = \int_{-\infty}^{\infty} \frac{-1}{y^2-2} dy \left(\text{put } x + \frac{1}{x} = y \right) = 0$$

$$\therefore I = \int_0^{\infty} \frac{dx}{1+x^4} = \int_0^{\infty} \frac{x^2 dx}{1+x^4}$$

$$2I = \int_0^{\infty} \frac{1+x^2}{1+x^4} dx = \int_{-\infty}^{\infty} \frac{dy}{y^2+2} = \frac{\pi}{\sqrt{2}} \left(\text{Put } x - \frac{1}{x} = y \right)$$

$$= \frac{\pi}{\sqrt{2}}$$

6. A, C

Sol. AC

$$f'(x) = 0 \Rightarrow (\sin x + \cos x)(\text{non zero quantity}) = 0$$

$$\Rightarrow \tan x = -1 \Rightarrow x = 3\pi/4 \text{ or } 7\pi/4.$$

$$\text{Global Min} = x = 2n\pi + (3\pi/4); \text{ global max} = x = 2n\pi + (7\pi/4)$$

$$M = \frac{4}{8-\sqrt{2}}; \quad m = \frac{4}{8+\sqrt{2}}$$

7. A, B, C

Sol. $y = x^{\log_x e} = e$ (constant) $\Rightarrow$ A and C

as $\lim_{x \rightarrow 1} f(x) = e$

8. C

Sol. $I = \int_0^1 \frac{\tan^{-1} x}{x} dx$ $x = \tan \theta; dx = \sec^2 \theta d\theta$

$$I = \int_0^{\pi/4} \frac{\theta \cdot \sec^2 \theta}{\tan \theta} d\theta = \int_0^{\pi/4} \frac{\theta}{\sin \theta \cos \theta} d\theta = \int_0^{\pi/4} \frac{2\theta}{\sin 2\theta} d\theta; \quad 2\theta = y$$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{y}{\sin y} dy = \frac{1}{2} \int_0^{\pi/2} \frac{x}{\sin x} dx$$

9. D

Sol. $f'(x) = 1 + \ln^2 x + 2 \ln x = 0 \Rightarrow (1 + \ln x)^2 = 0 \Rightarrow \frac{1}{e}$

Hence $f\left(\frac{1}{e}\right) = 1 + \frac{1}{e} + \int_1^{\frac{1}{e}} (\ln^2 t + 2 \ln t) dt = 1 + \frac{1}{e} + t \ln^2 t \Big|_1^{\frac{1}{e}} = 1 + \frac{1}{e} + \frac{1}{e} = 1 + 2e^{-1}$

10. C

Sol. $f(x) = \frac{t+3x-x^2}{x-4}; \quad f'(x) = \frac{(x-4)(3-2x) - (t+3x-x^2)}{(x-4)^2}$

for maximum or minimum, $f'(x) = 0$

$$-2x^2 + 11x - 12 - t - 3x + x^2 = 0$$

$$-x^2 + 8x - (12+t) = 0$$

For one M and m,

$$D > 0$$

$$64 - 4(12+t) > 0$$

$$16 - 12 - t > 0 \Rightarrow 4 > t \quad \text{or } t < 4$$

11. A

Sol. $h(x) = \ln(f(x) \cdot g(x)) = \ln e^{\{y\} + [y]} = \{y\} + [y] = y = e^{|x|} \operatorname{sgn} x$

$$\therefore h(x) = e^{|x|} \operatorname{sgn} x = \begin{cases} e^x & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -e^{-x} & \text{if } x < 0 \end{cases}$$

$$h(-x) = \begin{cases} e^{-x} & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ -e^x & \text{if } x > 0 \end{cases}$$

$$h(x) + h(-x) = 0 \text{ for all } x$$

12. A

$$\int e^x \left[\tan^{-1} x + \frac{2x}{(1+x^2)^2} \right] dx$$

$$= \int e^x \left[\tan^{-1} x - \frac{1}{1+x^2} + \frac{1}{1+x^2} + \frac{2x}{(1+x^2)^2} \right] dx$$

$$\text{Let } f(x) = \tan^{-1} x - \frac{1}{1+x^2}$$

[Step-2]:

$$f'(x) = \frac{1}{1+x^2} + \frac{1(2x)}{(1+x^2)^2}$$

$$= \int e^x [f(x) + f'(x)] dx$$

$$= e^x f(x) + C$$

$$= e^x \left[\tan^{-1} x - \frac{1}{1+x^2} \right] + C$$

13. C

Sol. $f(2^+) = 8; \quad f(2^-) = 16$

14. C

Sol. Given $\frac{d^2 y}{dx^2} + y \frac{dy}{dx} = 0$

$$\text{now } \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} \therefore \frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{\frac{dx}{dy}} \right) = \frac{d}{dy} \left(\frac{1}{\frac{dx}{dy}} \right) \cdot \frac{dy}{dx} = - \frac{1}{\left(\frac{dx}{dy} \right)^2} \cdot \frac{d^2 x}{dy^2} \cdot \frac{1}{\frac{dx}{dy}}$$

$$\frac{d^2 y}{dx^2} = - \frac{\frac{d^2 x}{dy^2}}{\left(\frac{dx}{dy} \right)^3} \quad (\text{putting in (1)})$$

$$- \frac{\frac{d^2 x}{dy^2}}{\left(\frac{dx}{dy} \right)^3} + y \frac{dy}{dx} = 0 \Rightarrow y \left(\frac{dx}{dy} \right)^2 - \frac{d^2 x}{dy^2} = 0 \Rightarrow C$$

15. A

Sol. As $m=9>0$, hence we can substitute $\sqrt{9x^2+4x+6} = u \pm 3x$.

16. D

Sol. Here as per notations given, we can substitute $\sqrt{1+x^2} = (u-x)$ as $m=1>0$ and $p=1>0$.

$$\Rightarrow I = \int \frac{u^{15}}{u} du = \int u^{14} du = \frac{1}{15} u^{15} + c = \frac{1}{15} \left(x + \sqrt{1+x^2} \right)^{15} + c.$$

17. B

Sol. $f(x) = e^{\lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} - 1 \right) \cdot n}$

$$\text{Substituting } n = \frac{1}{t^2}, \text{ we get } f(x) = e^{\lim_{t \rightarrow 0} \left(\frac{\cos tx - 1}{t^2} \right)} = e^{-\frac{x^2}{2}}$$

18. D

Sol. $g(x) = -x^{4b}$, $b = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1} \right) = \lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1 - x^2 - 1}{\sqrt{x^2 + x + 1} + \sqrt{x^2 + 1}} \right) = \frac{1}{2}$

$\therefore g(x) = -x^2$